

Data Engineering Mini Project – Training Brief

Project Objective

This mini project is designed to assess freshers on core Data Engineering concepts using Databricks and Apache Spark. Participants will work on a real-world e-commerce dataset with multiple interrelated tables, apply data cleaning and transformation logic, implement a Medallion Architecture (Bronze, Silver, Gold), and build analytical fact and dimension tables along with business KPIs.

Dataset Used

Brazilian E-Commerce Public Dataset by Olist (Kaggle). This dataset represents orders placed on the Olist e-commerce platform and contains multiple related tables covering customers, orders, products, sellers, payments, reviews, and logistics.

Source Tables (Minimum to Use)

- 1 olist_orders_dataset
- 2 olist_order_items_dataset
- 3 olist_products_dataset
- 4 olist_customers_dataset
- 5 olist_sellers_dataset
- 6 olist_order_payments_dataset
- 7 olist_order_reviews_dataset
- 8 olist_geolocation_dataset

Architecture Requirements (Mandatory)

- 1 **Bronze Layer:** Raw ingestion of all CSV files into Delta tables without transformation. Schema inference must be avoided.
- 2 **Silver Layer:** Data cleaning, deduplication, data type standardization, null handling, date normalization, and referential integrity checks.
- 3 **Gold Layer:** Creation of business-ready fact and dimension tables and pre-aggregated KPI tables.

Data Quality & Cleaning Expectations

- 1 Remove duplicate order items and reviews
- 2 Handle missing customer city/state information
- 3 Standardize timestamp and date columns
- 4 Handle orders with multiple payments
- 5 Fix or flag products with missing category names
- 6 Validate delivery dates (delivered before purchase, late deliveries)
- 7 Ensure referential integrity between fact and dimension tables

Gold Layer – Data Model Expectations

Dimension Tables

- 1 dim_customer
- 2 dim_product
- 3 dim_seller
- 4 dim_date
- 5 dim_geography (city, state level)

Fact Tables

- 1 fact_orders
- 2 fact_order_items
- 3 fact_payments
- 4 fact_reviews

KPIs to be Built (Gold Layer Only)

- 1 Total revenue by month
- 2 Average Order Value (AOV)
- 3 Top 10 products by revenue
- 4 Top 10 sellers by revenue
- 5 Orders by customer city and state
- 6 New vs returning customers (monthly)
- 7 Customer Lifetime Value (CLV)
- 8 Order fulfillment rate
- 9 Late delivery percentage
- 10 Average delivery time (days)
- 11 Payment method distribution
- 12 Average review score by product category
- 13 Average review score by seller
- 14 Percentage of low-rated orders (rating < 3)

Evaluation Criteria

- 1 Correct implementation of Medallion Architecture
- 2 Data quality handling and robustness of transformations
- 3 Correct fact and dimension modeling
- 4 Efficient use of Spark APIs and SQL
- 5 Performance considerations (partitioning, joins)
- 6 Code readability, naming standards, and documentation

Deliverables

- 1 Databricks notebooks for Bronze, Silver, and Gold layers
- 2 Final Delta tables in the Gold layer
- 3 KPI tables built from Gold facts and dimensions
- 4 Project documentation (README or notebook markdown)

Optional Advanced Tasks (Bonus)

- 1 Incremental data loading logic
- 2 Slowly Changing Dimension (Type 1 or Type 2)
- 3 Data quality checks and reconciliation counts
- 4 Basic Databricks SQL or Power BI dashboard